To eliminate the effects of aliasing from this equation, imagine that the frequency domain is so finely sampled that it turns into a continuous curve. This makes the time domain infinitely long with no periodicity. The DTFT is the Fourier transform to use here, resulting in the time domain signal being given by the relation:

EQUATION 11-4
Inverse DTFT of the rectangular pulse. In the frequency domain, the pulse has an amplitude of one, and runs from zero frequency to the cutoff frequency, f_c, a value between 0 and 0.5. The time domain signal is held in $x[i]$ with i running from 0 to N-1. To avoid the division by zero, use $x[0] = 2f_c$.

$$x[i] = \frac{\sin(2\pi f_c i)}{i\pi}$$

This equation is very important in DSP, because the rectangular pulse in the frequency domain is the perfect *low-pass filter*. Therefore, the sinc function described by this equation is the filter kernel for the perfect low-pass filter. This is the basis for a very useful class of digital filters called the *windowed-sinc filters*, described in Chapter 15.

Figures (c) & (d) show that a triangular pulse in the time domain coincides with a sinc function *squared* (plus aliasing) in the frequency domain. This transform pair isn't as important as the *reason* it is true. A $2M - 1$ point triangle in the time domain can be formed by convolving an M point rectangular pulse with itself. Since convolution in the time domain results in multiplication in the frequency domain, convolving a waveform with itself will *square* the frequency spectrum.

Is there a waveform that is its own Fourier Transform? The answer is yes, and there is *only* one: the Gaussian. Figure (e) shows a Gaussian curve, and (f) shows the corresponding frequency spectrum, also a Gaussian curve. This relationship is only true if you ignore aliasing. The relationship between the standard deviation of the time domain and frequency domain is given by: $2\pi\sigma_f = 1/\sigma_t$. While only one side of a Gaussian is shown in (f), the negative frequencies in the spectrum complete the full curve, with the center of symmetry at zero frequency.

Figure (g) shows what can be called a **Gaussian burst**. It is formed by multiplying a sine wave by a Gaussian. For example, (g) is a sine wave multiplied by the same Gaussian shown in (e). The corresponding frequency domain is a Gaussian centered somewhere other than zero frequency. As before, this transform pair is not as important as the *reason* it is true. Since the time domain signal is the multiplication of two signals, the frequency domain will be the convolution of the two frequency spectra. The frequency spectrum of the sine wave is a delta function centered at the frequency of the sine wave. The frequency spectrum of a Gaussian is a Gaussian centered at zero frequency. Convolving the two produces a Gaussian centered at the frequency of the sine wave. This should look familiar; it is identical to the procedure of *amplitude modulation* described in the last chapter.